2025-spring Introduction to Quantum Information and Quantum Computing

HW # 1 (Due 3/30 midnight = 3/31 00:00 am). Hand in as a single pdf file through ETL. For Qiskit problems: Pasting only the result (output) of the code (perhaps as a picture) will suffice—no need to hand in the coding itself. Also, running on an actual backend is recommended if possible, but the simulation result is also Okay.

1. **Density Matrix.** Show that the purity of a density matrix is less than or equal to 1.

2. **Simple Tomography.** For two cases of a two-level system, (a) pure $\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)$ and (b) half and half mixture of $|0\rangle$ and $|1\rangle$, determine $\langle\sigma_X\rangle$, $\langle\sigma_Y\rangle$, and $\langle\sigma_Z\rangle$ and draw Bloch vector for each case on the Bloch sphere.

3. **Spin singlet and Triplet States. (20 points)** Alice and Bob each possess one member of a pair of interacting magnetic dipoles (spin-1/2 particles). The interaction Hamiltonian for two interacting magnetic dipoles separated by distance r is given by,

$$H_I = \frac{\mu^2}{r^3}(\sigma_{x,A}\otimes\sigma_{x,B}+\sigma_{y,A}\otimes\sigma_{y,B}-2\sigma_{z,A}\otimes\sigma_{z,B})$$

   A. Find the allowed energies of the system and show that the eigenvectors of the Hamiltonian include entangled states.

   B. Then, rewrite the Hamiltonian using its eigenvectors. And write down density matrices of eigenstates of this Hamiltonian.

4. **Greenberger-Horne-Zeilinger (GHZ) state.** For a system of n qubits, the GHZ state can be written as $|GHZ\rangle = \frac{|0\rangle^{\otimes n}+|1\rangle^{\otimes n}}{\sqrt{2}}$. Consider the simplest case $|GHZ\rangle_3 = \frac{|000\rangle+|111\rangle}{\sqrt{2}}$, and we call subsystems A, B, and C

   A. Show that the GHZ state is maximally entangled.

   B. Starting from $|GHZ\rangle_3 = \frac{|000\rangle+|111\rangle}{\sqrt{2}}$, trace out the subsystem A. Is the resultant state pure or mixed? If mixed, how much?

   C. Repeat the above problem (B) by tracing out the subsystem B and C.

5. **Qiskit - Generating and state tomography of the Bell states.** Using a Hadamard and controlled not gates, construct a quantum circuit that generates two-qubit entangled Bell states (all four of them). Paste (or draw yourself by hand) the circuit diagrams here. Then, perform state tomography and paste the result (e.g., a city map plot of the density matrices).

6. **Qiskit - Schmidt decomposition.** Use Qiskit – quantum info module that has a built in function for Schmidt decomposition for the following states. And then obtain all reduced density matrices.

   A. $\frac{|00\rangle+|01\rangle+|11\rangle}{\sqrt{3}}$ (A, B systems are both 2 dimensional)

   B. $\frac{2|00\rangle+|01\rangle-|10\rangle-2|11\rangle}{\sqrt{10}}$ (A, B systems are both 2 dimensional)

   C. $\frac{2|00\rangle+|01\rangle-|02\rangle-2|10\rangle+|11\rangle+|20\rangle+|22\rangle}{\sqrt{13}}$ (A, B systems are both 3 dimensional)

7. **Qiskit** –

   A. Generate a circuit that results the GHZ state given in Prob. 4. Perform state tomography and paste the result here.

   B. Challenge: Can you make GHZ state with more qubits, say, 20 qubits? See if you can do it with a real backend or simulation.